

SangWook Woo

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EDUCATION

B.S. in Electronic and Information Engineering, Soongsil University

- Period: Mar/2020 - Aug/2026 (Expected)
- GPA: 4.28 / 4.5 (144 credits completed)
- Major GPA: 4.25 / 4.5 (111 major credits completed)
- Two years leave of absence due to mandatory military service

Relevant Coursework

- Embedded / System: Embedded Systems, Microprocessor Applications, Digital FPGA Design, Operating System, Data Communication, Computer Architecture
- Circuit / Hardware: Digital System Design, Circuit Analysis, Electronic Circuits, Applied Electronic Circuits, Digital Signal Processing, Signals and Systems, Semiconductor Devices, Electromagnetics, Transmission and Propagation

AWARDS

- Hyeongnam Science Award - Bronze Prize, Soongsil University College of Engineering, Dec/2025
- [AIX2025] Deep Learning Hardware Design Competition - Encouragement Award, Seoul National University Next-Generation Semiconductor Research Group, Jun/2025
- 2021 Soongsil Capstone Design Competition - Encouragement Award, Soongsil University, Sep/2021

SCHOLARSHIP

- National Science & Engineering Scholarship, Mar/2024 - Sep/2025
- Heunghan Foundation Scholarship, Mar/2021 - Sep/2021
- Academic Excellence Scholarship (Quarter), Sep/2020

EXPERIENCE

Undergraduate Research Intern supervised by Prof. Incheol Song

- Period: Dec/2025 - Jan/2026
- Designed and implemented an AXI-to-APB bridge using Verilog-HDL
- Implemented burst-to-single transfer handling, address decoding, and APB slave selection
- Performed simulation and functional verification with Verilog testbenches across read/write scenarios

FPGA Video Filtering Accelerator Project

- Period: Sep/2025 - Nov/2025
- Built a real-time 3x3 video filtering system on Ultra96-V2 (Zynq) in an HW/SW co-design flow, streaming OV5640 camera input directly to a TFT-LCD
- Designed the Window3x3 and LCD/output-buffer control logic in Verilog, applying double buffering, CDC synchronization, and frame-drop based sync control
- Implemented PS-side C code for camera initialization and filter control, and verified real-time performance at 81.97 FPS after full-system integration

PCB Circuit Design & PSpice Designer Training

- Period: Jul/2025 (60 hours)
- Completed the PCB Circuit Design / Capture / PSpice Designer training at 9ES Campus (Cadence Authorized Training)
- Learned PCB schematic design, simulation, and analysis using Capture and PSpice Designer
- Gained hands-on experience in PCB layout and circuit implementation through project-based training
- Applied simulation techniques for verifying and optimizing PCB designs

Microprocessor Applications

- Period: Mar/2025 - Jun/2025
- Studied ARM processor architecture
- Served as team leader for an image processing project
- Optimized image processing performance with ARM Assembly

Operating System Project

- Period: Mar/2025 - Jun/2025
- Solved system-level tasks in the xv6 operating system
- Learned core operating system concepts including virtualization, concurrency, and persistence

System Programming Project

- Period: Mar/2025 - Jun/2025
- Developed a SIC/XE assembler with C and Java
- Developed a SIC/XE simulator in Java to support instruction execution and memory management

AGV Path Planning and Deadlock Resolution Simulation

- Period: Mar/2025 - Nov/2025
- Implemented a control algorithm to reduce collision and deadlock issues among multiple AGVs in an unmanned parking environment
- Combined WHCA*, WFG/SCC-based deadlock detection, and Partial CBS-based replanning logic
- Compared performance with A and D Lite on realistic and extended parking maps, validating throughput and computational efficiency in dense environments
- This project led to the Hyeongnam Science Award - Bronze Prize

[AIX2025] Deep Learning Hardware Design Competition

- Period: Feb/2025 - May/2025
- Studied quantization algorithms, CNNs, FPGA resource utilization, Vivado, and AXI protocols
- Designed and verified a custom CNN accelerator in Verilog, optimizing for limited FPGA resources and high-throughput performance
- Analyzed related research papers to better understand and implement the AI accelerator design process

Embedded System Project

- Period: Sep/2024 - Jan/2025
- Gained hands-on experience with ATtiny and STM32 microcontrollers using Microchip Studio and STM32CubeIDE
- Controlled various peripherals including sensors, switches, LEDs, motors, and touch panels using C, HAL libraries, and direct register access
- Analyzed datasheets and designed efficient circuits under limited pin constraints

- Implemented interrupts, timers, and communication protocols such as SPI and I2C

Undergraduate Research Intern supervised by Prof. Incheol Song

- Period: Dec/2024 - Jan/2025
- Designed and implemented the AES algorithm using Verilog-HDL
- Developed RTL-level hardware design and integrated an APB interface for register-level communication
- Performed simulation and functional verification using Verilog testbenches

Semiconductor and Display Engineering Class Project

- Period: Nov/2024 - Dec/2024
- Studied semiconductor processes for one semester
- Target circuit: LDO (Low-dropout regulator)
- Designed the circuit layout to minimize area while reflecting design rule constraints
- Planned an optimized fabrication process

Small-sized Elevator Control System Project

- Period: Sep/2024 - Dec/2024
- Served as team leader
- Designed and implemented a miniature elevator system using the STM32F103
- Programmed motor control, button input, floor display, and floor queue management logic in C

Digital System Design Class Project

- Period: Oct/2024 - Nov/2024
- Served as team leader
- Developed FIR filter architecture using Verilog-HDL at the RTL level
- Coordinated task division, integration, and simulation using testbenches
- Received an A+ in the course

Full-Custom IC Design Course

- Period: Jul/2024 - Oct/2024 (60 hours)
- Completed the Cadence Full-Custom IC Design & Term Project course at Soongsil University
- Learned Cadence Virtuoso, Schematic, Layout, Spectre, and Assura
- Designed basic logic gates and a 4-bit full adder/subtractor using full-custom layout and schematic
- Performed DRC/LVS checks and circuit simulation with Spectre

Soongsil Capstone Design Competition

- Period: Jun/2021 - Sep/2021
- Developed a COVID-19 prevention automatic door system
- Utilized Arduino and Raspberry Pi for sensor control and automation
- Implemented mask detection, thermal temperature screening, and CCTV monitoring using cameras and infrared sensors

Soongsil Electronic Exhibition (SEE) Club

- Period: Mar/2021 - Present
- Participated in the annual SEE exhibition project
- Instructed new SEE members in C programming
- Provided academic and embedded systems project mentoring to junior members

CERTIFICATION

- OPIc IH, Mar/2025 - Mar/2027
- Computer Specialist in Spreadsheet & Database Level-1, Jul/2023 - Present
- Korean Driver's License (Class 2), Dec/2022 - Present
- Craftsman Information Processing, May/2019 - Present

SKILLS

Language

- Verilog-HDL | Intermediate: Used continuously in major projects, undergraduate internships, and competitions. Built experience in RTL design and testbench-based verification through AXI-to-APB interfaces, AES-128 encryption modules, FIR filters, and FPGA video filtering projects.
- C | Intermediate: Used as a primary language in MCU control, operating systems coursework, and embedded projects. Implemented sensor, motor, and LCD control in STM32 and AVR-based projects.
- ARM Assembly | Intermediate: Able to convert simple C functions into Assembly and optimize them. Applied it directly to improve arithmetic performance in image processing projects.
- SystemVerilog | Basic: Understand SystemVerilog syntax and the concepts of class-based verification and coverage. Used it in simple design and verification tasks.
- Python | Basic: Can use Python for simple automation, data processing, and helper scripts.
- Java | Basic: Used in a system programming course for a SIC/XE assembler simulator. Understand basic syntax and can write simple code.

Tool

- Vivado | Intermediate: Used continuously in FPGA projects for synthesis, implementation, simulation result analysis, and resource utilization review. Also explored Tcl and batch-script-based automation flows.
- STM32CubeIDE / Microchip Studio | Intermediate: Used for MCU control, debugging, peripheral control, and embedded development environment setup in STM32 and AVR-based projects.
- Linux Environment | Intermediate: Used as a development environment for projects and coursework. Comfortable with basic commands, file management, and build environment setup.
- Git | Intermediate: Used not only for version control in personal projects but also for branch-based collaboration and history management in team projects.
- Vitis | Basic: Used in the FPGA Video Filtering project for PS-side driver code and AXI4-Lite based control implementation.
- Visual Studio | Basic: Used as a development environment for C/C++ assignments and projects.
- Cadence Xcelium | Basic: Used in digital circuit design and verification tasks.
- Keil | Basic: Used in ARM Assembly based image processing projects for coding, debugging, and performance comparison.
- Virtuosio / Capture / PCB Editor | Basic: Used in short courses and training programs for circuit design, simulation, PCB layout, and basic verification.
- Design Compiler | Basic: Learned the basic flow, commands, constraint application, and test procedures in a semiconductor design training course.
- MS Office (Word, Excel) | Intermediate: Comfortable with basic document work and data organization.

Others

- Hands-on experience with oscilloscopes and pulse generators

- Hands-on experience with soldering PCB boards and measuring/validating hardware